Guest Lecture: Data Handling

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November 27, 2025

Outline

1. Introduction
2. Practical Knowledge and Experience in Research & Financial Sector
3. Exercise

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1 Introduction

2 Practical Knowledge and Experience in Research & Financial Sector

3 Exercise

1. Introduction: About myself

Short CV:

- 24-Current: Macroeconomic research analyst, BIS
- 24: Off-cycle interest rate research analyst, Goldman Sachs
- 23-24: Research assistant in Quantitative methods, HSG
- 20-22: Economist, BoC(HK)/ Swiss Re
- 17-20: Treasury analyst/ Money market dealer, CCB(Asia)

Education:

- 22-24: MA in Quantitative Economics & Finance (MiQE/F), HSG
- 13-17: BBA in Economics & Finance, HKUST

1. Introduction: Take-Home Messages

For Programming Newbies:

- Programming skills are increasingly necessary in financial sector!
- Fake it till you make it (don't be afraid)!

1. Introduction: Key Take-Home Messages

For Everyone:

- Practice makes perfect
- Better documentation with a solid understanding of each step

It goes a long way!

1. Introduction: Key Take-Home Messages

On Chat-GPT/ Co-pilot:

- It helps significantly. But you need to act as the conductor/ maestro!
- Understand → Guide → Outcome → Check & Verify

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3. Exercise

2. Financial sector: Pricing data & formula

Treasury analyst/ Money market dealer, CCB(Asia):

Financial instrument & Price:

- Short-term money lending/ borrowing, repo, FX forward, FX swap
- Buy/ Sell? What's your bid/ offer? Flow trade/ Prop. trade?

Mark-to-Market (MTM) and P&L:

- Data: Funding cost, risk ratios, forward point, discount rate etc
- (unrealized) P&L reconciliation

3) $F = S_0 \left(\frac{1 + \left[r_q \times \left(\frac{d}{360} \right) \right]}{1 + \left[r_b \times \left(\frac{d}{360} \right) \right]} \right)$

F = *Forward Price*
S_0 = *Spot Price*
r_q = *qoute currency interest rate*
r_b = *base currency interest rate*
T = *Time to maturity. I assume would be* $\left(\frac{d}{360} \right)$

FX forward fair value

$$\mathbf{PV} = \sum_i \mathbf{CF}_i \times D(0, t_i)$$

Present Value (PV)

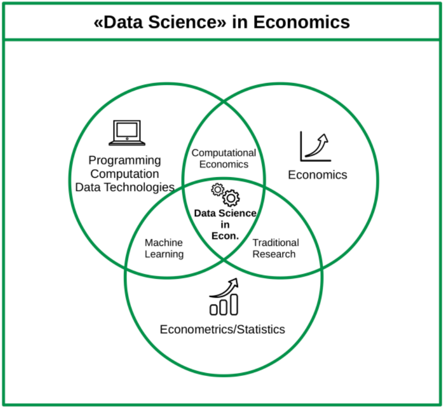
2. Financial sector: Pricing data & formula

Treasury analyst/ Money market dealer, CCB(Asia):

Not much "Data Science" / "Quant", but you need to...:

- understand the products, business and data very well
- follow the market closely

Yet having some programming skills can really help save time!



2. Academic Research: Apply theory to practice

Research assistant in Quantitative methods, HSG:

Improve model forecasting performance:

- New framework: Multi-frequency echo state network (MS-ESN)

$$\mathbf{x}_t = F(\mathbf{x}_{t-1}, \mathbf{z}_t). \tag{1}$$

$$\mathbf{y}_{t+1} = h_\theta(\mathbf{x}_t) + \epsilon_t \tag{2}$$

- **z**: Input features (eg IP, Retail sales etc); **x**: Reservoir states/ Neurons; **y**: GDP (output/ target)

Hyperparameter in play:

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Reservoir computing for macroeconomic forecasting with mixed-frequency data☆

Giovanni Ballarin [a], Petros Dellaportas [b,c], Lyudmila Grigoryeva [d,e,*], Marcel Hirt [h,1], Sophie van Huellen [f,g], Juan-Pablo Ortega [h]

In the case of an ESN model, the state and observation equations (2.1)–(2.2) are given by

$$\boldsymbol{x}_t = \alpha \boldsymbol{x}_{t-1} + (1-\alpha)\,\sigma(A\boldsymbol{x}_{t-1} + C\boldsymbol{z}_t + \boldsymbol{\zeta}), \tag{2.3}$$

$$\boldsymbol{y}_{t+1} = \boldsymbol{a} + W^\top \boldsymbol{x}_t + \boldsymbol{\epsilon}_t, \tag{2.4}$$

where $A \in \mathbb{M}_N$ is the reservoir matrix, $C \in \mathbb{M}_{N,K}$ is the input matrix, $\boldsymbol{\zeta} \in \mathbb{R}^N$ is the input shift, $\alpha \in [0, 1)$ is the leak rate, and $W \in \mathbb{M}_{N,J}$ denotes the readout coefficients. The map $\sigma : \mathbb{R} \to \mathbb{R}$ is an activation function applied elementwise, which in what follows we take to be the hyperbolic tangent. We refer to A, C, $\boldsymbol{\zeta}$ as state parameters that are randomly generated. Notice that if $A = 0$ and $\alpha = 0$, the state equation reduces to a nonlinear regression model with random coefficients (or a feedforward neural

2. Academic Research: Apply theory to practice

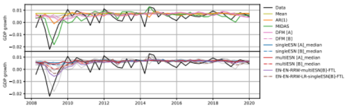
Research assistant in Quantitative methods, HSG:

From Many Models, One: Macroeconomic Forecasting with Reservoir Ensembles

Giovanni Ballarin [1] Lyudmila Grigoryeva [1,2] Yui Ching Li [3]

[1]University of St. Gallen [2]University of Warwick [3]Bank for International Settlements (BIS)

Macroeconomic (Multi-Frequency) Forecasting



Many different mixed-frequency forecasting/nowcasting approaches have been developed (Andreou et al., 2011; Modugno, 2013; Kim and Swanson, 2018; Huber et al., 2023).

Multi-frequency setting: **Low-frequency** target time series, $Y_t \in \mathbb{R}$; **High-frequency** regressors, $\{Z_s^{(q)}\}_{s \in \mathbb{Z}}$, $Z_s^{(q)} \in \mathbb{R}^{d_q}$, $q \in \{1, \ldots, Q\}$ with $Q \in \mathbb{N}$ groups of time series (e.g. grouped by frequency). Here, $\{\kappa_1, \ldots, \kappa_Q\}$ are the frequencies and s is a high-frequency time index.

Reservoir-Based Nonlinear State Space Approach

Ballarin et al. (2024) propose a new model class based on Echo State Networks – a *reservoir computing* approach (Gonon et al., 2020) – such that the state parameters are random.

Echo State Networks:

$$X_t = \alpha X_{t-1} + (1-\alpha)\sigma(\rho \overline{A} X_{t-1} + \gamma \overline{C} Z_t + \varsigma \overline{\zeta}),$$
$$Y_{t+1} = \eta + W^\top X_t + \epsilon_{t+1}.$$

- X_t: Reservoir state.
- $\overline{A}, \overline{C}, \overline{\zeta}$: Normalized reservoir parameters, randomly sampled.
- $\alpha, \rho, \gamma, \varsigma$: Hyperparameters.
- W: Trainable coefficients.

Multi-Frequency ESNs (MFESNs): (Ballarin et al., 2024) For each group, $q \in \{1, \ldots, Q\}$, (e.g. specific frequency) introduce

$$X_s^{(q)} = \alpha_q X_{s-1}^{(q)} + (1-\alpha_q)\sigma(\rho_q \overline{A}_q X_{s-1}^{(q)} + \gamma_q \overline{C}_q Z_s^{(q)} + \varsigma_q \overline{\zeta}_q).$$

The regression equation then combines (frequency) groups:

$$Y_{t+1} = \eta_Q + \textstyle\sum_{q=1}^{Q} W_q^\top X_{t(s)}^{(q)} + \epsilon_{t+1}.$$

MFESNs can achieve **state-of-the-art forecasting performance** for quartely GDP forecasting at multiple horizons, and can also be used for nowcasting.

Theoretical Results

We extend known regret bounds (c.f. Mourtada and Gaïffas, 2019) to the **dependent case**.

Assumption 1: Losses $\ell_t^{(k)}$ are such that $\ell_t^{(k)} \in [0,1]$ a.s. for all $k \in \{1, \ldots, K\}$, $t \in \{1, \ldots, T\}$.

Assumption 2: Losses $\ell_t^{(k)}$ are random variables such that either (i) $\{\ell_t^{(k)}\}$ are i.i.d. across $t \in \{1, \ldots, T\}$, or (ii) $\{\ell_t^{(k)}\}$ have uniformly mixing coefficients $\varphi_t^{(k)}$.

Assumption 3: For all $k \in \{1, \ldots, K\}$, $\{\ell_t^{(k)}\}$ is a weakly stationary process with $\mu_k = \mathbb{E}[\ell_t^{(k)}]$.

Theorem. Let the *sub-optimality gap* be $\Delta := \min_{k \neq k^*} \mu_k - \mu_{k^*}$, where $\mu_k = \mathbb{E}[\ell_t^{(k)}]$ and $k^* := \arg\min_{k \in [K]} \mu_k$. Suppose that almost surely there are no ties in the FTL weights for all $t \geqslant 1$: (i) If Assumption 2(i) holds, then $\mathbb{E}[\overline{R}_{\mathrm{FTL},T}] \leqslant 2 + (2\log(K) + 4)/\Delta^2$; (ii) If Assumption 2(ii) holds and uniform mixing coeffs. are such that $\overline{\theta} = \max_{k \in [K]} \sup_{t \geqslant 1} \theta_t^{(k)} < \infty$, then

$$\mathbb{E}[\overline{R}_{\mathrm{FTL},T}] \leqslant 3 + \frac{8\overline{\theta}(\log(K) + 4)}{\Delta^2}.$$

Empirical Results – U.S. GDP Growth Forecasting

Data: *medium-MD* dataset of Ballarin et al. (2024), combining FRED-QD (quarterly), FRED-MD (monthly), and Refinitiv Datastream (daily) data sources:
→ 30 total regressors, available from January 1st, 1990, to December 31st, 2019

Models: Benchmarks are in-sample mean and AR(1) model of U.S. GDP growth. Baselines are the exact MFESN models (same random draws) of Ballarin et al. (2024). Ensembles are constructed with simple averaging, rolling MSE, FTL, Hedge, and Adaptive Hedge (AdaHedge) schemes. Testing and online learning sample ranges from 2008Q1 to 2019Q4.

Forecasting Results

Model	Baseline	Median	Ensemble: Average	RollMSE	FTL	Hedge	AdaHedge
Mean	1.000	-	-	-	-	-	-
AR(1)	0.758	-	-	-	-	-	-
EN-RP-ESN (random parameters resampling)							
S-MFESN A	0.969 -	0.940 *-2.94%*	0.918 *-5.24%*	0.915 *-5.56%*	**0.664** *-31.49%*	0.915 *-5.53%*	0.716 *-26.07%*
S-MFESN B	0.815 -	0.887 *+8.75%*	0.815 *+7.57%*	0.875 *+7.28%*	**0.619** *-24.08%*	0.875 *+7.31%*	0.679 *-16.69%*
M-MFESN A	0.901 -	0.895 *-0.71%*	0.889 *-1.38%*	0.887 *-1.52%*	**0.731** *-18.89%*	0.887 *-1.51%*	0.750 *-16.70%*
M-MFESN B	0.682 -	0.738 *+8.18%*	0.730 *+7.08%*	0.727 *+6.60%*	**0.587** *-13.87%*	0.727 *+6.61%*	0.598 *-12.28%*
EN-αRP-ESN (random parameters resampling & different leak rates)							

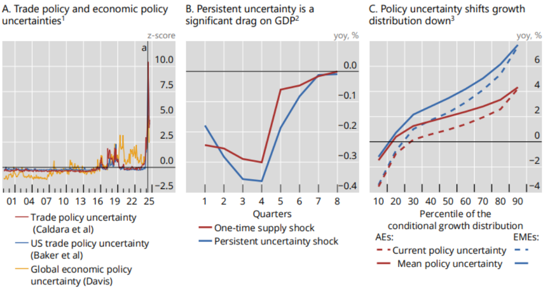
2. Macro Research: A little bit of everything

Macroeconomic research analyst, BIS:

Data analysis, modeling, visualization etc:

- Link: BIS Bulletin

Trade policy uncertainty weighs on growth — Graph 3



[a] 2 April 2025: "Liberation Day".

[1] Z-score from 2000 to present. [2] Impulse response functions of GDP to shocks identified through sign restrictions (see Burgert et al (2025)). "One-time supply shock" is a one-time one standard deviation shock that pushes up inflation while lowering investment and GDP growth. "Persistent uncertainty shock" is a one standard deviation shock, spread out over three quarters, that raises uncertainty while lowering investment and GDP growth. [3] Expected growth rate within four quarters, based on a growth-at-risk model that features the current growth rate, one- and 10-year government bond yields, the deviation of the real effective exchange rate from each country's historical average, economic policy uncertainty, the VIX index and country dummies. Economic policy uncertainty is standardised country by country. The model was estimated based on 21 countries in a 2000–25 sample period. For AEs, GDP-weighted average for CA, EA, GB, JP and US; for EMEs GDP-weighted average for BR, CL, IN and KR.

Sources: Baker et al (2016); Burgert et al (2025); Caldara et al (2020); Davis (2016); Bloomberg; LSEG Datastream; BIS.

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3. Exercise: A glimpse into my work

Off-cycle interest rate research analyst, Goldman Sachs
Economist, BoC(HK)/ Swiss Re:

Outright 10y US Treasury:

- Is it a good time to increase allocations for long-term investors?
- What risks lie ahead?

Key Points to Address:

- Real yield relative high compared to historical period (since 2000)
- Risks: Rising US inflation and increasing debt-to-GDP ratio

3. Exercise: A glimpse into my work

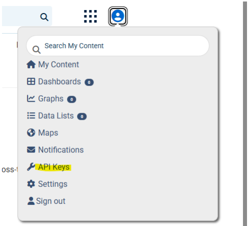
Key Points to Address:

- Real yield relative high compared to historical period (since 2000)
- Risks: Rising US inflation and increasing debt-to-GDP ratio

Data required (all from FRED) → Tasks:

- Real yield from 2000 (DFII10) → Show the current real yield percentile
- US recent CPI numbers (CPIAUCSL) → Show inflation (yoy, %) from 2024
- US public debt-to-GDP ratio (GFDEGDQ188S) → Show line chart from 2015

Link: Register for a FRED account



3. Exercise: A taste of my task

Link: Register for a FRED account

```r
# Load the relevant library
library(fredr)
# Set your FRED API key to access the Federal Reserve Econ
fredr_set_key("XXX")
```

Figure 1: Load FRED library and input API Key

```r
# Import US 10y real yield
data <- fredr(
  series_id = "DFII10",   # Market Yield on U.S. Treasury
  observation_start = as.Date("2000-01-01"),  # Start dat
  observation_end = as.Date("2025-11-30")     # End date
)

# Check the summary of data
summary(data)
```

Figure 2: Get data for real yield

```
> summary(data)
      date              series_id             value          realtime_start        realtime_end
 Min.   :2003-01-02   Length:5968        Min.   :-1.1900   Min.   :2025-11-18   Min.   :2025-11-18
 1st Qu.:2008-09-21   Class :character   1st Qu.: 0.2700   1st Qu.:2025-11-18   1st Qu.:2025-11-18
 Median :2014-06-10   Mode  :character   Median : 0.9500   Median :2025-11-18   Median :2025-11-18
 Mean   :2014-06-10                      Mean   : 0.9676   Mean   :2025-11-18   Mean   :2025-11-18
 3rd Qu.:2020-02-27                      3rd Qu.: 1.8450   3rd Qu.:2025-11-18   3rd Qu.:2025-11-18
 Max.   :2025-11-17                      Max.   : 3.1500   Max.   :2025-11-18   Max.   :2025-11-18
                                         NA's   :245
```

Figure 3: Data summary

3. Exercise: A taste of my task - Conclusion